DEPARTMENT OF BIO-STRUCTURAL ENGINEERING

NEURAL SYSTEMS, SPINAL DECOMPRESSION & CRITICAL OVERRIDES

MANUAL CODE: KLIMEK-NCLEX // AUTHORIZED PERSONNEL ONLY

NotebookLM

INTERFACE LEGEND // SYSTEM DIAGNOSTICS
STABLE
EXPECTED
MONITOR
CAUTION
CRITICAL
DANGER (LEVEL D)
INTERVENTION
ACTION
MECHANISM
PATHOPHYSIOLOGY

NotebookLM

DIN Condensed

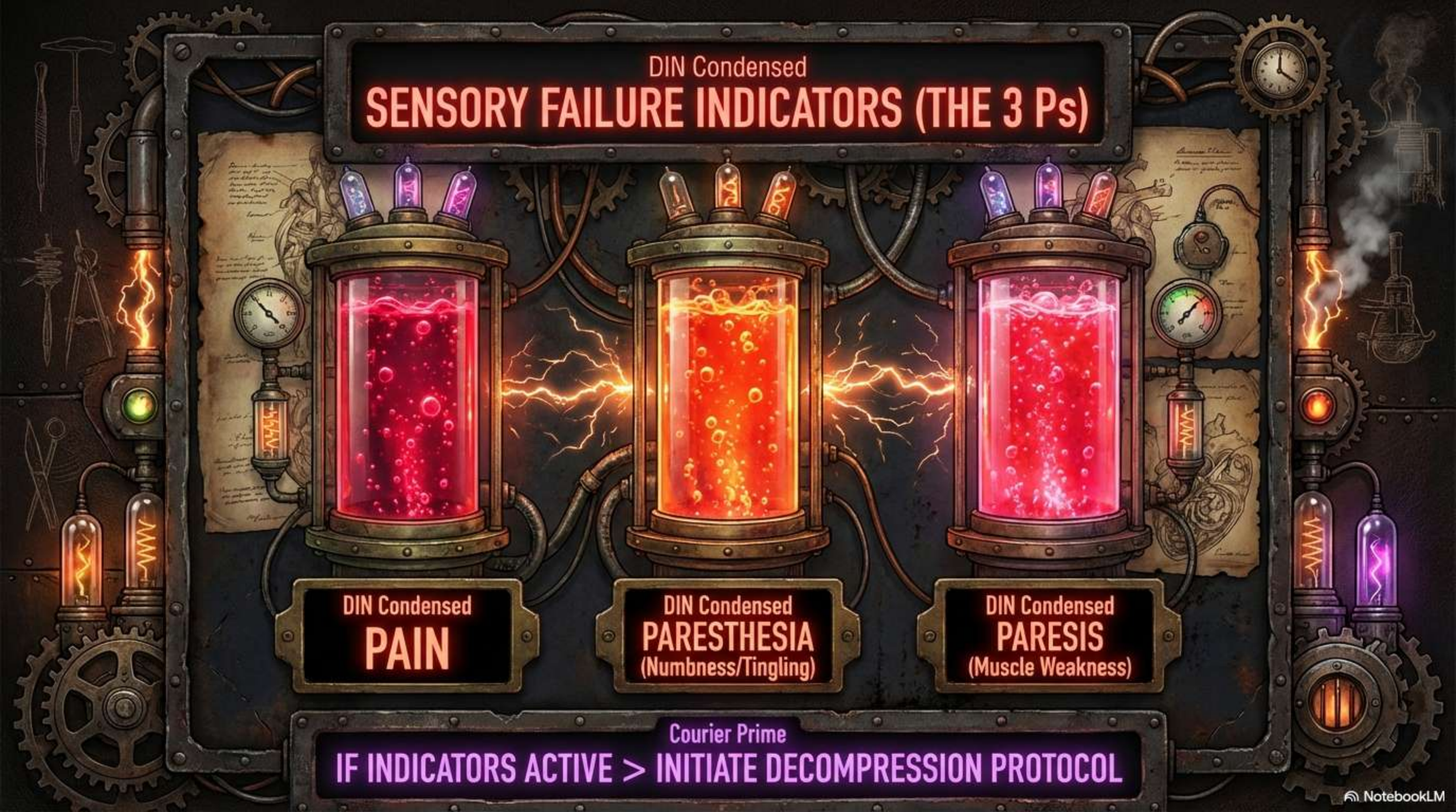
STRUCTURAL DECOMPRESSION: LAMINECTOMY

DEFINITION: Ectomy = Removal. Lamina = Vertebral Spinous Processes (Wing Bones).
OBJECTIVE: To relieve nerve root compression.
VISUALIZATION: Ventilation of the Nerve Root.

NotebookLM

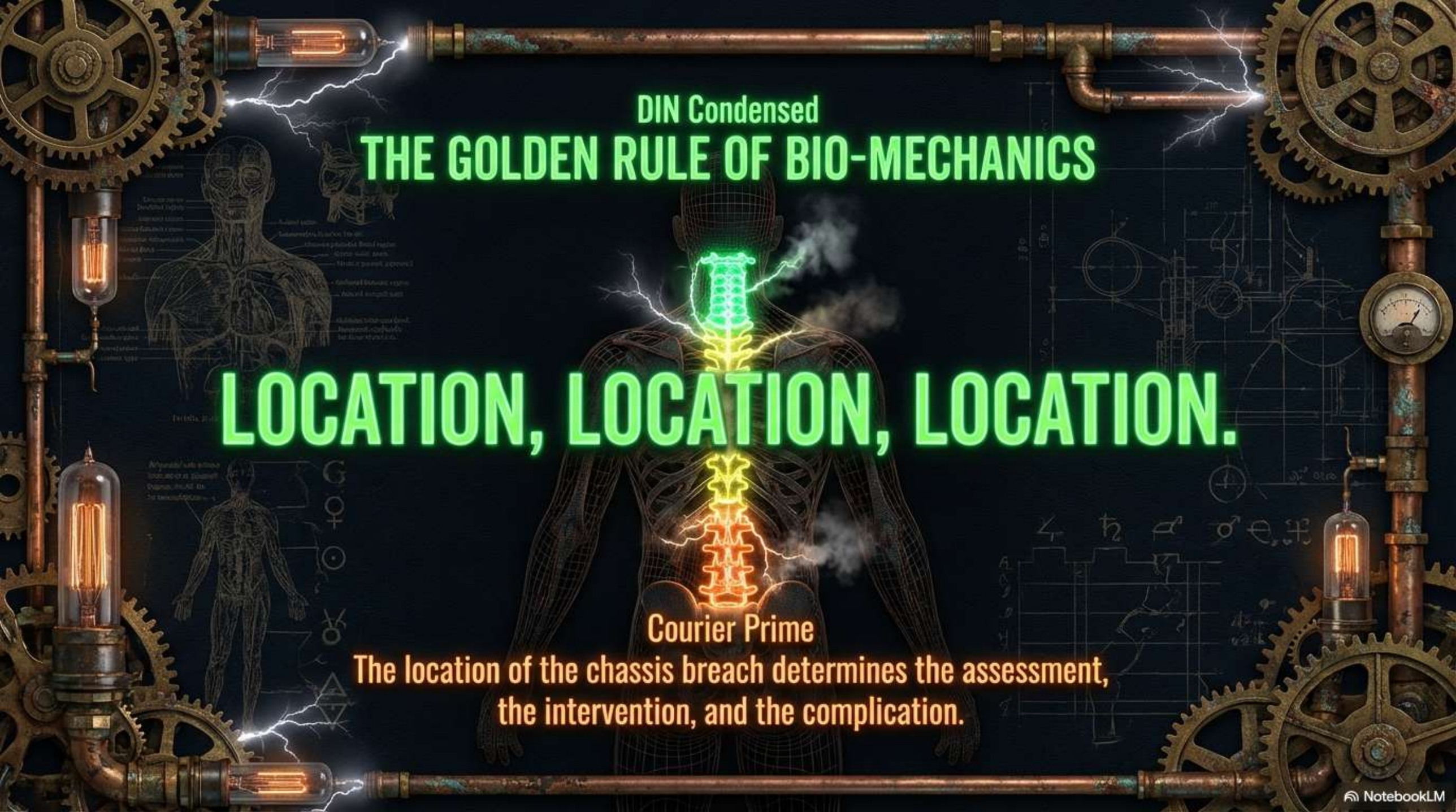
DIN Condensed

SENSORY FAILURE INDICATORS (THE 3 Ps)



DIN Condensed

THE GOLDEN RULE OF BIO-MECHANICS



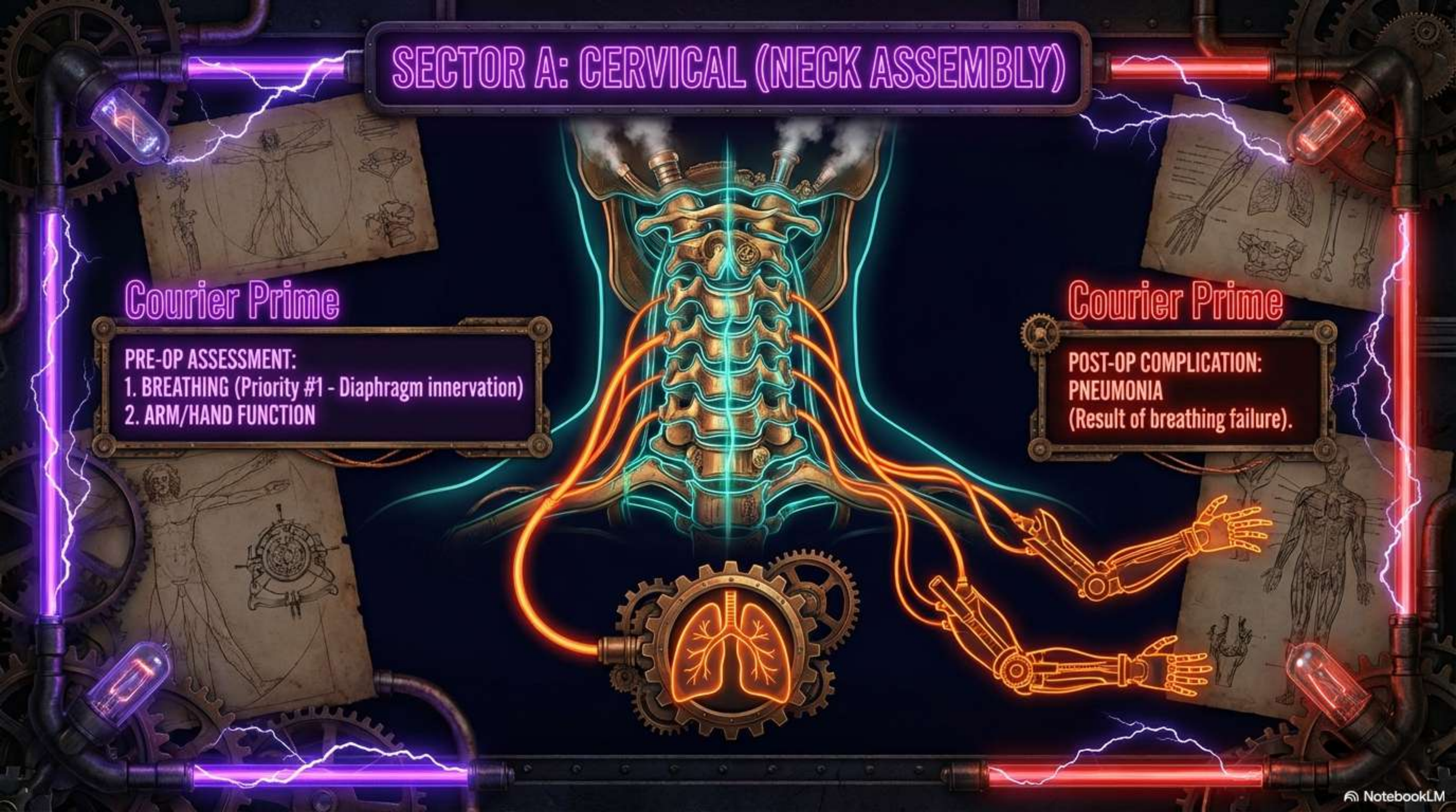
LOCATION, LOCATION, LOCATION.

Courier Prime

The location of the chassis breach determines the assessment, the intervention, and the complication.

NotebookLM

SECTOR A: CERVICAL (NECK ASSEMBLY)



NotebookLM

SECTOR B: THORACIC (UPPER BACK)

Courier Prime

PRE-OP ASSESSMENT:
1. COUGH MECHANISM (Needs abdominal contraction)
2. BOWEL SOUNDS

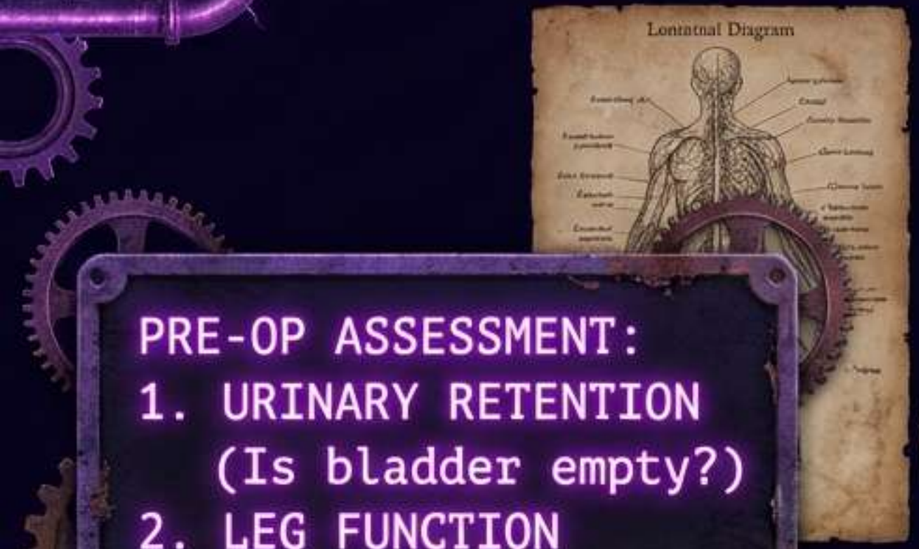
Courier Prime

POST-OP COMPLICATIONS:
1. PNEUMONIA
   (Inability to clear airway)
2. PARALYTIC ILEUS (Gut Stasis)

NotebookLM

SECTOR C: LUMBAR (LOWER BACK)

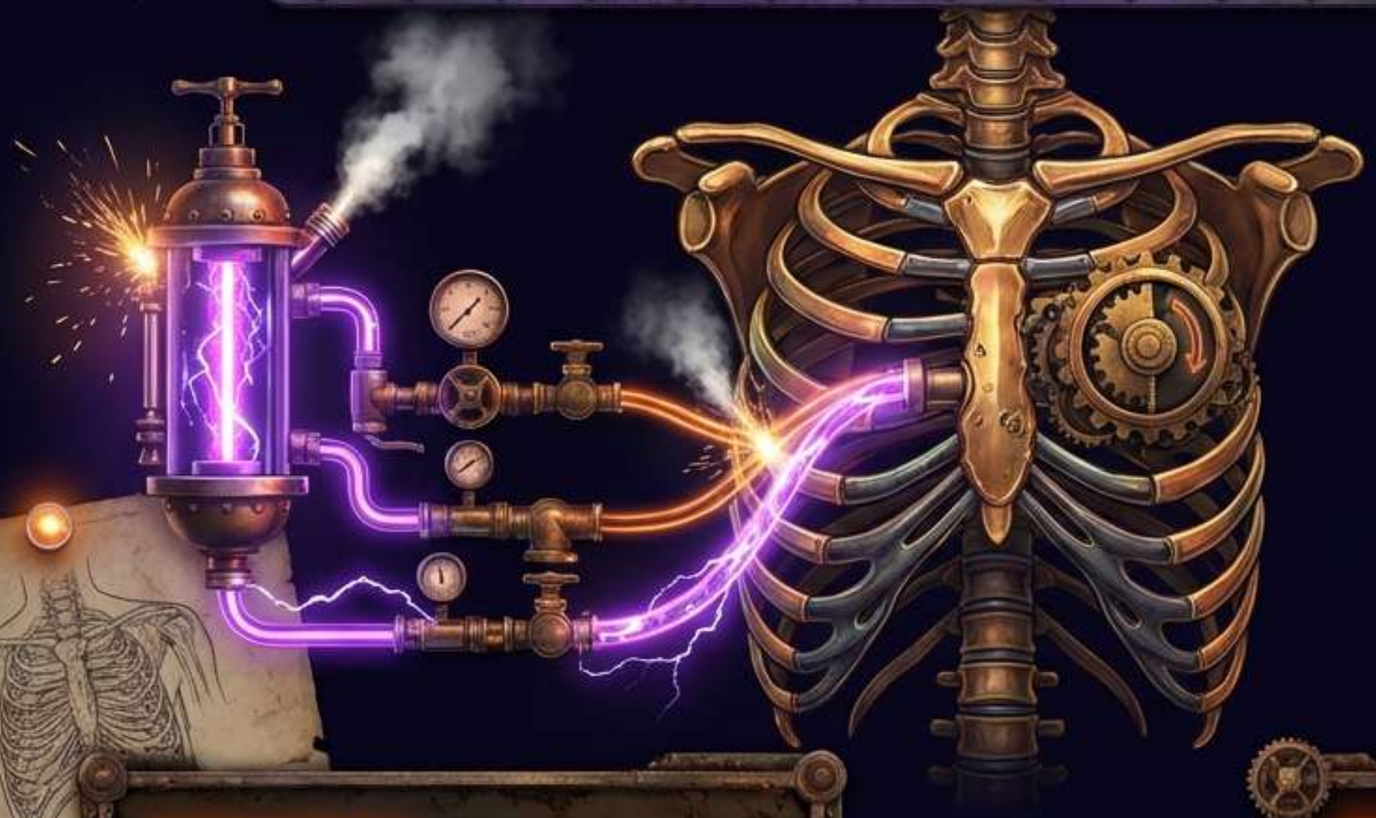
PRE-OP ASSESSMENT:
1. URINARY RETENTION
   (Is bladder empty?)
2. LEG FUNCTION



POST-OP COMPLICATIONS:
1. URINARY RETENTION
2. LEG PARESIS

NotebookLM

SPECIAL CONFIGURATION: ANTERIOR THORACIC



REQUIREMENT:
CHEST TUBE
(Pneumothorax management).

DEFINITION:
Surgery performed from the front, through the chest cavity to the spine.

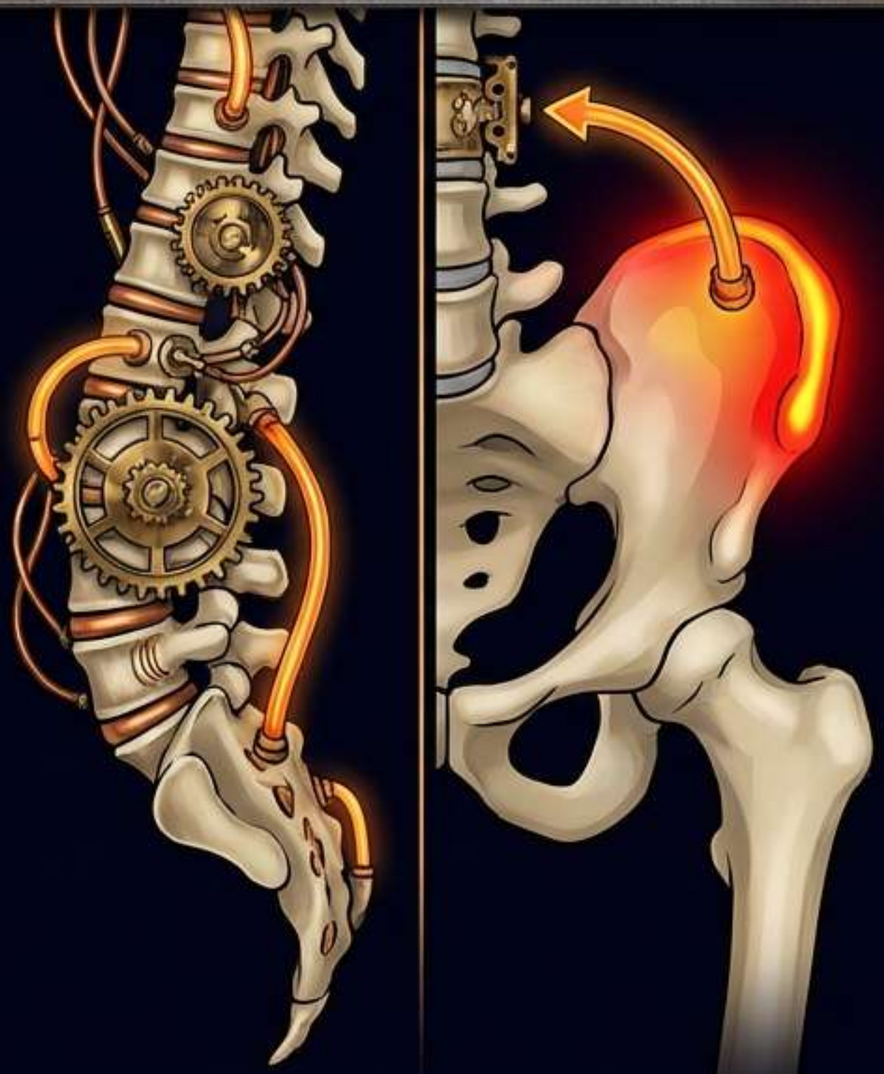
NOTE:
Only Anterior Thoracic laminectomies require chest tubes.

NotebookLM

LAMINECTOMY WITH FUSION

MECHANISM:
Bone graft taken from Iliac Crest (Hip) to bridge the spinal gap.

OBJECTIVE:
Structural stability.
Eliminate bone-on-bone grinding.

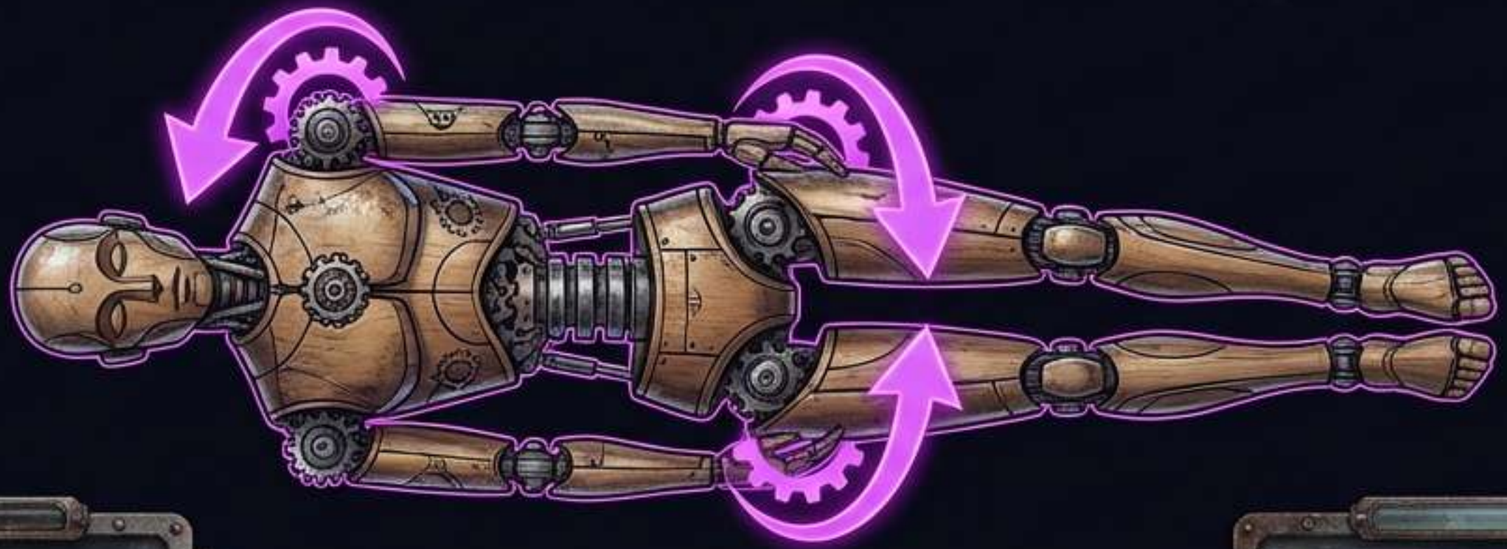


INCISION 1 (HIP):
Most Pain.
Most Bleeding/Drainage (JP/Hemovac).
Risk of Infection.

INCISION 2 (SPINE):
Highest Risk of Rejection.

NotebookLM

TRANSPORT PROTOCOL #1: LOG ROLL



COMMAND:
LOG ROLL.

EXECUTION:
The chassis must move as a solid unit. No twisting. No torque.

APPLICATION:
Mandatory for ALL spinal surgeries and injuries.

NotebookLM

MOBILIZATION TIMERS

Courier Prime
RULE 1: Do NOT dangle the legs (Orthostatic risk).

RULE 2: SIT < 30 MINUTES. (Sitting places max pressure on intake valves).

Courier Prime
PERMITTED ACTIONS:
Walk, Stand, Lie Down. (Unlimited duration).

NotebookLM

DIN Condensed

DISCHARGE ORDERS: TEMPORARY RESTRICTIONS

Courier Prime
DURATION: 6 WEEKS.

1. Do not sit > 30 mins.
2. Lie flat / Log roll.
3. No driving.
4. Lift nothing > 5 lbs (Gallon of Milk).

NotebookLM

DISCHARGE ORDERS: PERMANENT RESTRICTIONS

DURATION: FOREVER.

1. Never lift by bending at waist (Use Knees).
2. Cervical Lams: Lift nothing over head.
3. No jerky rides (Rollercoasters, Horseback, Off-road biking).

NotebookLM

AMBULATORY MECHANICS: CRUTCHES

MEASUREMENT PROTOCOL:

1. Length: 2-3 finger widths below anterior axillary fold to a point lateral/front of foot.
2. Hand Grip: Elbow flexion ~30 degrees.

RISK: Nerve damage if resting on axilla.

NotebookLM

AMBULATORY MECHANICS: GAIT LOGIC

RULE: Even for Even, Odd for Odd.

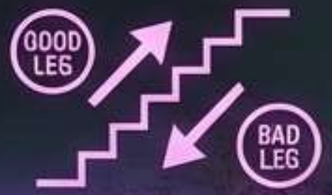
1. 2-POINT: Move crutch + opposite foot. (Mild weakness).

2. 3-POINT: Move 2 crutches + bad leg. (One leg affected).

3. 4-POINT: Move everything separately. (Severe bilateral weakness).

4. SWING-THROUGH: For Amputees / Non-weight bearing.

5. STAIRS: UP with the GOOD, DOWN with the BAD.



NotebookLM

CATASTROPHIC SYSTEM FAILURE: SCI

Courier Prime

CERVICAL INJURY:
Quadriplegia (Full Chassis Failure).

Courier Prime

THORACIC/LUMBAR INJURY:
Paraplegia (Lower Chassis Failure).

Courier Prime

NOTE: Airway assessment remains Priority #1 for Cervical.

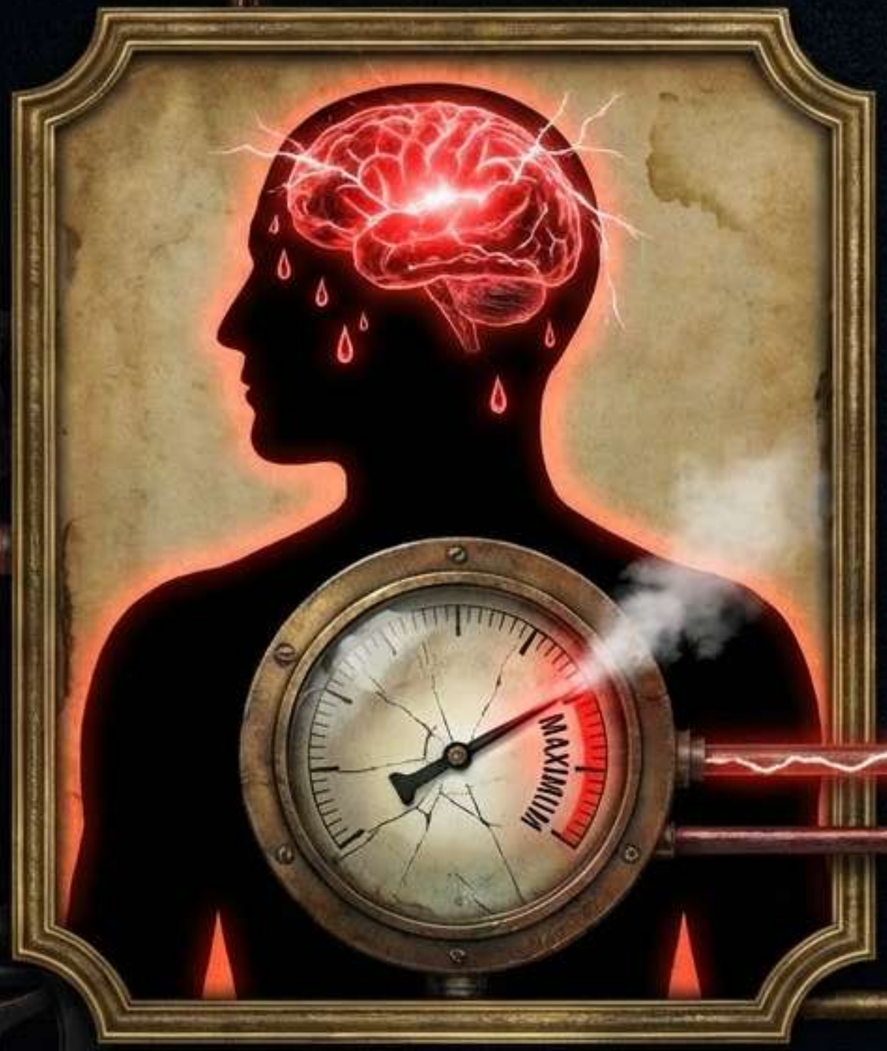
NotebookLM

CRITICAL OVERPRESSURE: AUTONOMIC DYSREFLEXIA
Courier Prime
TARGET SYSTEMS: Spinal Cord Injuries at T6 or higher.
STATUS: MEDICAL EMERGENCY. Priority Level D.
TRIGGER: Noxious stimuli (Full Bladder / Impacted Bowel / Tight Clothing).

NotebookLM

DIN Condensed

DYSREFLEXIA GAUGE READINGS (S/S)



Courier Prime

PRESSURE: SEVERE HYPERTENSION (Stroke RIsk).

HEAD: Pounding Headache.

HEART: Bradycardia.

SKIN: Profuse SWEATING and Gooseflesh *above* the level of injury.

NotebookLM

DIN Condensed

EMERGENCY VENTING SEQUENCE

Courier Prime

STEP 1: RAISE HOB (High Fowler's).
*First Action (Gravity Assist).

STEP 2: CHECK BLADDER
(Distention? Kinked Foley? Catheterize).

STEP 3: CHECK BOWEL
(Impaction?)

STEP 4: LOOSEN CLOTHING.

STEP 5: ANTI-HYPERTENSIVES
(If mechanical venting fails).

NotebookLM

DIN Condensed

CONSCIOUSNESS METER (GCS)

Courier Prime

SCALE: 3 (Deep Coma) to 15 (Fully Alert).

CRITICAL THRESHOLD: LESS THAN 8 = INTUBATE.

COMPONENTS: Eye Opening, Verbal Response, Motor Response.

NotebookLM